

- 1- The HTTP Request-Response Cycle is the process through which client (usually a web browser) communicate with a web server to request & receive web resources such as HTML, CSS, Java script, Images or data.

Steps

- User enters a URL
- Browser sends an HTTP request
- Server processes the request
- Server sends a HTTP response
- Browser receives and displays the webpage

2- Web Technologies

- HTML

- HTML - Creates the structure of webpage.
- Features - Headings, paragraphs, Images.

- CSS - designs and styles webpages.
- Features - colors, fonts, layouts.

- JavaScript - Adds interactivity to webpages.
- Features - Events, calculations, validation.

- Bootstrap - is a CSS framework for creating responsive websites quickly.
- Feature - Grid system, button, cards, forms.

- JQuery - is a Javascript library that makes Javascript easier to use.
- Features - DOM manipulation, events, ...

3 - Client-Side

- Code that runs in the user's browser.
- HTML, CSS, JAVASCRIPT.
- UI, animations, form validation & user interaction

★ → Server-Side

- Code that runs on the web server.
- Node.js, PHP, Python, Java.
- Database operations, authentication & business logic.

- Client-Side

- Runs in browser
- Handles UI
- HTML, CSS, JS
- User can usually see the code

Server-Side

- Runs on Server
- Handles data / business logic
- PHP, Python, JAVA, Node.js
- Server code is generally hidden

- In Simple Words:

- Client-Side = Browser ~~for~~ ~~Room~~
- Server-Side = Server ~~for~~ ~~Room~~

4 - Block-level elements start on a new line & usually take the full available width

→ Characteristics

- Start on a new line.
- Take full width available.
- Can contain other elements

eg - `<div>`, `<p>`, `<h1>`, `<section>`

5 - Semantic HTML elements clearly describe the meaning & purpose of the content.

→ Importance

- Makes code easier to understand
- Improves SEO
- Helps accessibility
- Makes webpage structure clear

eg - `<header>`, `<nav>`, `<main>`, `<section>`

6 `<table border="1">`

`<tr>`

`<th> Ser. No. </th>`

`<th> Name </th>`

`<th> Designation </th>`

`<th> Department </th>`

`</tr>`

```
<tr>
  <td> 1 </td>
  <td> John </td>
  <td> Manager </td>
  <td> Sale </td>
</tr>
```

```
<tr>
  <td> 2 </td>
  <td> Smith </td>
  <td> Sec. Manager </td>
  <td> Marketing </td>
</tr>
```

```
<tr>
  <td> 3 </td>
  <td> Jones </td>
  <td> Manager </td>
  <td> HR </td>
</tr>
```

```
</table>
```

```
<ul>
```

```
<li> River
  <ul>
    <li> Congo </li>
    <li> Amazon </li>
    <li> Brahmaputra </li>
    <li> Macmado </li>
  </ul>
```

```
</li>
</ul>
```

8- Types of CSS

- 1- Inline CSS - Applied directly to an HTML element.
- 2- Internal CSS - written inside the <Style> tag
- 3- External CSS - written in a separate CSS file.

9- HTML vs CSS vs Java Script

- 1★ HTML → Structure & content
- 2★ CSS → Presentation & Design
- 3★ JAVAScript → Behavior & Interactivity

1★ Structural framing

2★ Interior Design, ~~front~~ paint & Decoration

3★ Electrical wiring, plum

1★ Defines elements like headings, paragraphs, Image

2★ controls colors, fonts, layout, responsiveness.

3★ Handles dynamic contents, Interactive elements.

10 - Role of Bootstrap

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- Bootstrap is a CSS framework used to create responsive websites quickly. It provides ready-made buttons, cards, forms.
- JQuery - It is a JavaScript library that makes tasks like DOM manipulation, events, animations easier.
- ReactJS - It is a JavaScript library used to create interactive user interfaces using reusable components.